

Activity - Estimating Lifetime

From `6_regression.ipynb` : build a Kaplan-Meier survival analysis on the Telco churn data and compare lifetime between payment methods.

A Kaplan-Meier estimate needs exactly three things - how long each subject was observed, whether the observation ended in the event or was cut short, and which group the subject belongs to. That is what picks the columns out of the 21 in the file:

survival term	Telco column	why
duration T	<code>tenure</code>	months as a customer, the only column measuring elapsed time
event E	<code>Churn == "Yes"</code>	whether that time ended in churn, or is still running
group	<code>PaymentMethod</code>	what the activity asks to compare
-	<code>customerID</code>	not a variable, just a check that one row means one customer

The other seventeen columns are customer attributes. None of them enter a Kaplan-Meier curve, which knows only about time, event and group; `Contract` comes back later when the Cox model needs something to hold constant. `TotalCharges` stays out on purpose even then - a running total of billing is mostly a restatement of how long someone has been paying, which is the very thing being estimated.

```
In [1]: from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from lifelines import KaplanMeierFitter
from lifelines.statistics import multivariate_logrank_test, pairwise_logrank_test
from lifelines.utils import restricted_mean_survival_time

pd.set_option("display.precision", 3)

df = pd.read_csv("Telco-Churn.csv")
df["T"] = df["tenure"].astype(float)
df["E"] = (df["Churn"] == "Yes").astype(int)

WORKING = ["customerID", "PaymentMethod", "tenure", "Churn", "T", "E"]
```

```
print(df.shape)
df[WORKING].head()
```

(7043, 23)

```
Out[1]:
```

	customerID	PaymentMethod	tenure	Churn	T	E
0	7590-VHVEG	Electronic check	1	No	1.0	0
1	5575-GNVDE	Mailed check	34	No	34.0	0
2	3668-QPYBK	Mailed check	2	Yes	2.0	1
3	7795-CFOCW	Bank transfer (automatic)	45	No	45.0	0
4	9237-HQITU	Electronic check	2	Yes	2.0	1

```
In [2]: # T is the x axis of every curve in this notebook, so its largest value is the
point
# past which there is no data - nothing can be said about month 73 or beyond.
print(f"T: {df['T'].min():.0f} to {df['T'].max():.0f} months,
{df['T'].nunique()} distinct values")
print(f"one row per customer: {df['customerID'].nunique()} ids across {len(df)}
rows")
```

T: 0 to 72 months, 73 distinct values
one row per customer: 7043 ids across 7043 rows

So every survival curve below stops at **month 72** - a six-year window with no month missing inside it, and no customer counted twice. Read nothing past that edge: the estimates end where the data does.

The other end, month 0, is the one that needs a decision.

1. What is in the data

```
In [3]: raw = pd.read_csv("Telco-Churn.csv")
raw.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines           7043 non-null   object
8   InternetService        7043 non-null   object
9   OnlineSecurity          7043 non-null   object
10  OnlineBackup            7043 non-null   object
11  DeviceProtection       7043 non-null   object
12  TechSupport            7043 non-null   object
13  StreamingTV            7043 non-null   object
14  StreamingMovies         7043 non-null   object
15  Contract               7043 non-null   object
16  PaperlessBilling        7043 non-null   object
17  PaymentMethod           7043 non-null   object
18  MonthlyCharges          7043 non-null   float64
19  TotalCharges            7043 non-null   object
20  Churn                   7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

```

Two things to take from that. Nothing is null, so there is no imputation to argue about. And `TotalCharges` came in as `object` rather than a number - eleven rows hold a blank string instead of an amount, and those eleven are exactly the customers with `tenure` zero, who signed up too recently to have been billed. The next section drops them.

2. Censoring

The point of Kaplan-Meier: most rows are *censored*, not observed failures. Averaging `tenure` would treat "still here after 60 months" as a lifetime of 60, which understates every group.

```

In [4]: print(f"customers          {len(df)}")
        print(f"churned (events)    {df.E.sum()}  ({df.E.mean():.1%})")
        print(f"still active        {(1 - df.E).sum()}  (right-censored)")
        print(f"tenure == 0                {(df['T'] == 0).sum()}  (brand new, no follow-up time)")
        print()

        # A zero-length observation carries no information for the estimator and
        # lifelines
        # warns about it, so those rows come out.

```

```
df = df[df["T"] > 0].copy()
print(f"kept {len(df)} rows")
```

```
customers          7043
churned (events)    1869  (26.5%)
still active        5174  (right-censored)
tenure == 0         11  (brand new, no follow-up time)
```

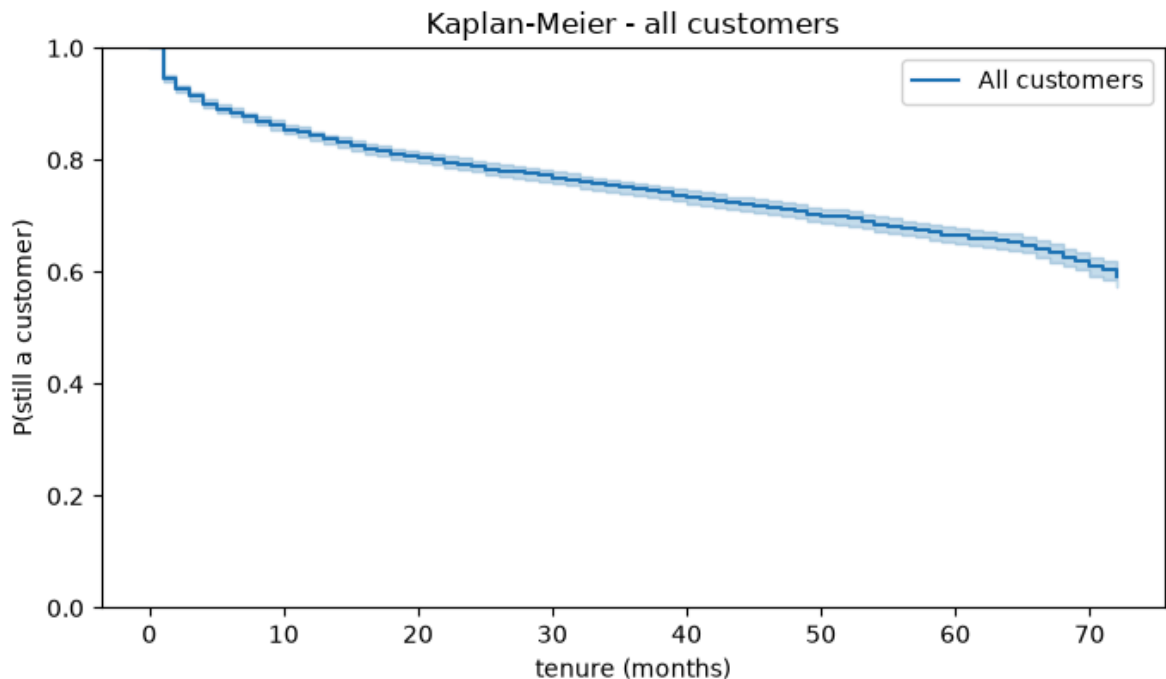
kept 7032 rows

3. Overall survival curve

```
In [5]: kmf = KaplanMeierFitter()
kmf.fit(df["T"], event_observed=df["E"], label="All customers")

ax = kmf.plot_survival_function(figsize=(7, 4.2))
ax.set(xlabel="tenure (months)", ylabel="P(still a customer)",
       title="Kaplan-Meier - all customers", ylim=(0, 1))
plt.tight_layout()
plt.show()

print("median survival time:", kmf.median_survival_time_)
print(kmf.survival_function_at_times([12, 24, 48, 72]).round(3))
```



```
median survival time: inf
12    0.843
24    0.789
48    0.709
72    0.593
Name: All customers, dtype: float64
```

4. By payment method

```

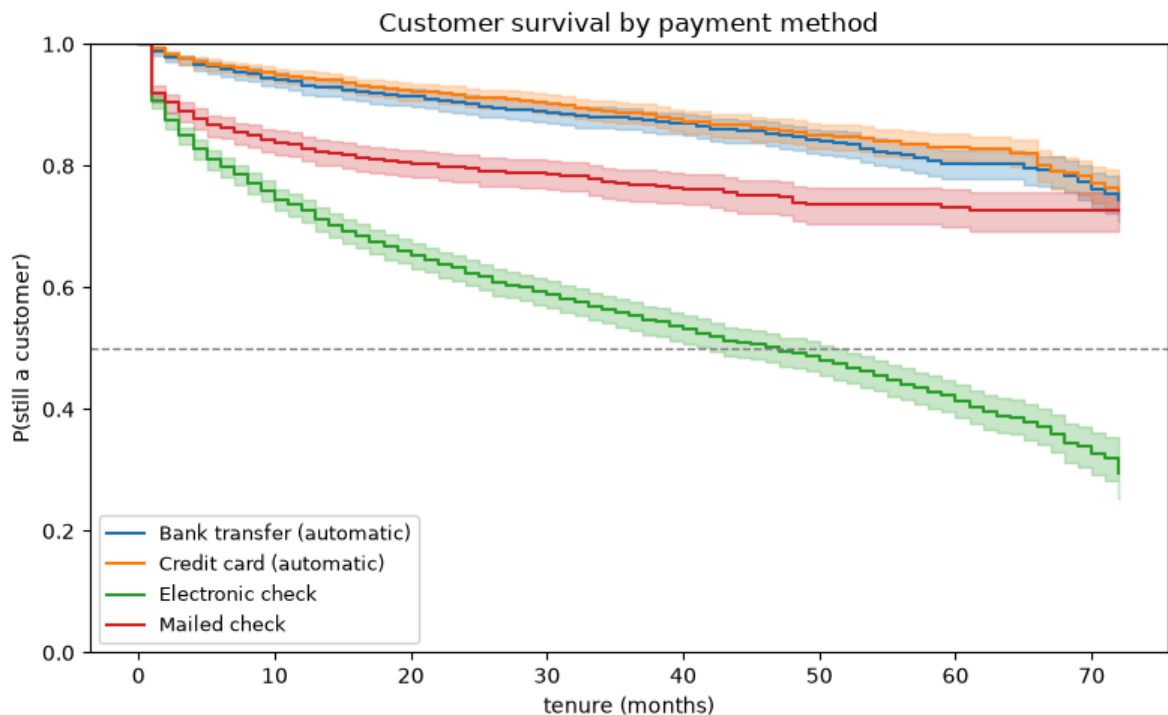
In [6]: METHODS = sorted(df["PaymentMethod"].unique())
fitters = {}

fig, ax = plt.subplots(figsize=(8, 5))
for method in METHODS:
    group = df[df["PaymentMethod"] == method]
    f = KaplanMeierFitter().fit(group["T"], event_observed=group["E"],
    label=method)
    fitters[method] = f
    f.plot_survival_function(ax=ax, ci_show=True)

ax.set(xlabel="tenure (months)", ylabel="P(still a customer)",
       title="Customer survival by payment method", ylim=(0, 1))
ax.axhline(0.5, color="grey", lw=1, ls="--")
ax.legend(loc="lower left", fontsize=9)

plt.tight_layout()
Path("figures").mkdir(exist_ok=True)
fig.savefig("figures/km_by_payment_method.png", dpi=150, bbox_inches="tight")
plt.show()

```



5. Comparing lifetime

`median_survival_time_` is the statistic the lecture uses, but three of the four curves never reach 0.5 inside the 72 months observed, so it comes back as `inf` for them. Two statistics that do exist for every group:

- **survival at fixed times** - the share still subscribed at 12, 24, 48, 72 months

- **RMST** (restricted mean survival time) - the area under the curve up to a chosen horizon, read as "expected months as a customer during the first N"

RMST needs that horizon picked by hand, and the choice is not neutral: over a short window everybody survives most of it, so the groups look closer together. Reporting several horizons keeps that visible. Twelve months is the one to quote - the risk set is still 55-87% full there, where at 72 months it is down to 1-10% and the curve is carried by a handful of customers.

```
In [7]: HORIZONS = [12, 24, 48, 72]
HEADLINE = 12

rows = []
for method, f in fitters.items():
    s = f.survival_function_at_times(HORIZONS).to_numpy()
    row = {
        "n": int(f.event_observed.shape[0]),
        "churned": int(f.event_observed.sum()),
        "median": f.median_survival_time_,
    }
    row |= {f"S({h})": v for h, v in zip(HORIZONS, s)}
    row |= {f"RMST({h})": restricted_mean_survival_time(f, t=h) for h in
HORIZONS}
    rows.append({"PaymentMethod": method} | row)

summary =
pd.DataFrame(rows).set_index("PaymentMethod").sort_values(f"RMST({HEADLINE})")
summary
```

Out[7]:

	n	churned	median	S(12)	S(24)	S(48)	S(72)	RMST(12)	RMST(24)
PaymentMethod									
Electronic check	2365	1071	47.0	0.727	0.624	0.493	0.295	9.863	17.9
Mailed check	1604	308	inf	0.828	0.797	0.740	0.726	10.545	20.2
Bank transfer (automatic)	1542	258	inf	0.931	0.902	0.847	0.745	11.561	22.5
Credit card (automatic)	1521	232	inf	0.945	0.913	0.855	0.758	11.623	22.7

```
In [8]: # How much of each horizon does a customer of this group actually spend
subscribed?
share = pd.DataFrame({
    f"{h} mo": summary[f"RMST({h})"] / h for h in HORIZONS
})
share.loc["gap (best - worst)"] = share.max() - share.min()
share
```

Out[8]:

	12 mo	24 mo	48 mo	72 mo
PaymentMethod				
Electronic check	0.822	0.748	0.653	0.573
Mailed check	0.879	0.845	0.808	0.783
Bank transfer (automatic)	0.963	0.941	0.909	0.875
Credit card (automatic)	0.969	0.949	0.918	0.886
gap (best - worst)	0.147	0.201	0.265	0.313

6. Are the differences real?

The log-rank test compares the curves without assuming any shape for them.

```
In [9]: overall = multivariate_logrank_test(df["T"], df["PaymentMethod"], df["E"])
print(f"all four groups: chi2 = {overall.test_statistic:.1f}, p = {overall.p_value:.3g}")
print()

pairwise = pairwise_logrank_test(df["T"], df["PaymentMethod"], df["E"])
pairwise.summary[["test_statistic", "p"]].round(4)
```

all four groups: chi2 = 865.2, p = 3.07e-187

Out[9]:

		test_statistic	p
Bank transfer (automatic)	Credit card (automatic)	0.868	0.351
	Electronic check	510.035	0.000
	Mailed check	51.072	0.000
Credit card (automatic)	Electronic check	539.740	0.000
	Mailed check	64.815	0.000
Electronic check	Mailed check	152.456	0.000

7. Cox proportional hazards

The curves above compare payment methods without holding anything else constant, and payment method is not chosen at random - it travels with contract type.

```
In [10]: pd.crosstab(df["PaymentMethod"], df["Contract"], normalize="index").round(3)
```

Out[10]:

	Contract	Month-to-month	One year	Two year
PaymentMethod				
Bank transfer (automatic)		0.382	0.254	0.364
Credit card (automatic)		0.357	0.262	0.381
Electronic check		0.782	0.147	0.071
Mailed check		0.557	0.209	0.234

Electronic-check customers are on month-to-month contracts 78% of the time against 36% for credit card, so the lifetime gap above could be contract type in disguise. A Cox model holds contract constant and separates the two.

The coefficients are read as **hazard ratios**, `exp(coef)` : the multiplier on the instantaneous churn rate relative to the reference level, at every point in time. HR = 2 means twice the churn rate of the reference; HR = 1 means no difference.

Reference levels: credit card (automatic), two-year contract.

```
In [11]: from lifelines import CoxPHFitter
from lifelines.statistics import proportional_hazard_test

REFERENCE = ["PaymentMethod_Credit card (automatic)", "Contract_Two year"]

design = pd.get_dummies(df[["PaymentMethod", "Contract"]],
dtype=float).drop(columns=REFERENCE)
design["tenure"] = df["T"].to_numpy()
design["churned"] = df["E"].to_numpy()

payment_cols = [c for c in design.columns if c.startswith("PaymentMethod")]
design.head()
```

Out[11]:

	PaymentMethod_Bank transfer (automatic)	PaymentMethod_Electronic check	PaymentMethod_Mailed check	Contract_1 to
0	0.0	1.0	0.0	
1	0.0	0.0	1.0	
2	0.0	0.0	1.0	
3	1.0	0.0	0.0	
4	0.0	1.0	0.0	

Model A - payment method alone

```
In [12]: cox_a = CoxPHFitter().fit(design[payment_cols + ["tenure", "churned"]],
    "tenure", "churned")

    cox_a.summary[["exp(coef)", "exp(coef) lower 95%", "exp(coef) upper 95%", "p"]]
```

Out[12]:

	exp(coef)	exp(coef) lower 95%	exp(coef) upper 95%	p
covariate				
PaymentMethod_Bank transfer (automatic)	1.089	0.912	1.301	3.446e-01
PaymentMethod_Electronic check	4.621	4.005	5.332	1.130e-97
PaymentMethod_Mailed check	2.149	1.810	2.551	2.540e-18

Model B - payment method adjusted for contract type

```
In [13]: cox_b = CoxPHFitter().fit(design, "tenure", "churned")

    print(f"concordance model A {cox_a.concordance_index_:.3f}    model B  

    {cox_b.concordance_index_:.3f}")
    cox_b.summary[["exp(coef)", "exp(coef) lower 95%", "exp(coef) upper 95%", "p"]]
```

concordance model A 0.676 model B 0.809

Out[13]:

	exp(coef)	exp(coef) lower 95%	exp(coef) upper 95%	p
covariate				
PaymentMethod_Bank transfer (automatic)	1.028	0.861	1.228	7.593e-01
PaymentMethod_Electronic check	2.104	1.822	2.430	3.981e-24
PaymentMethod_Mailed check	1.848	1.555	2.197	3.447e-12
Contract_Month-to-month	54.102	39.712	73.706	3.566e-141
Contract_One year	6.888	4.944	9.598	4.068e-30

```
In [14]: comparison = pd.DataFrame({
    "HR alone (A)": cox_a.hazard_ratios_,
    "HR adjusted (B)": cox_b.hazard_ratios_[payment_cols],
})
    comparison["shrinkage"] = 1 - comparison["HR adjusted (B)"] / comparison["HR
```

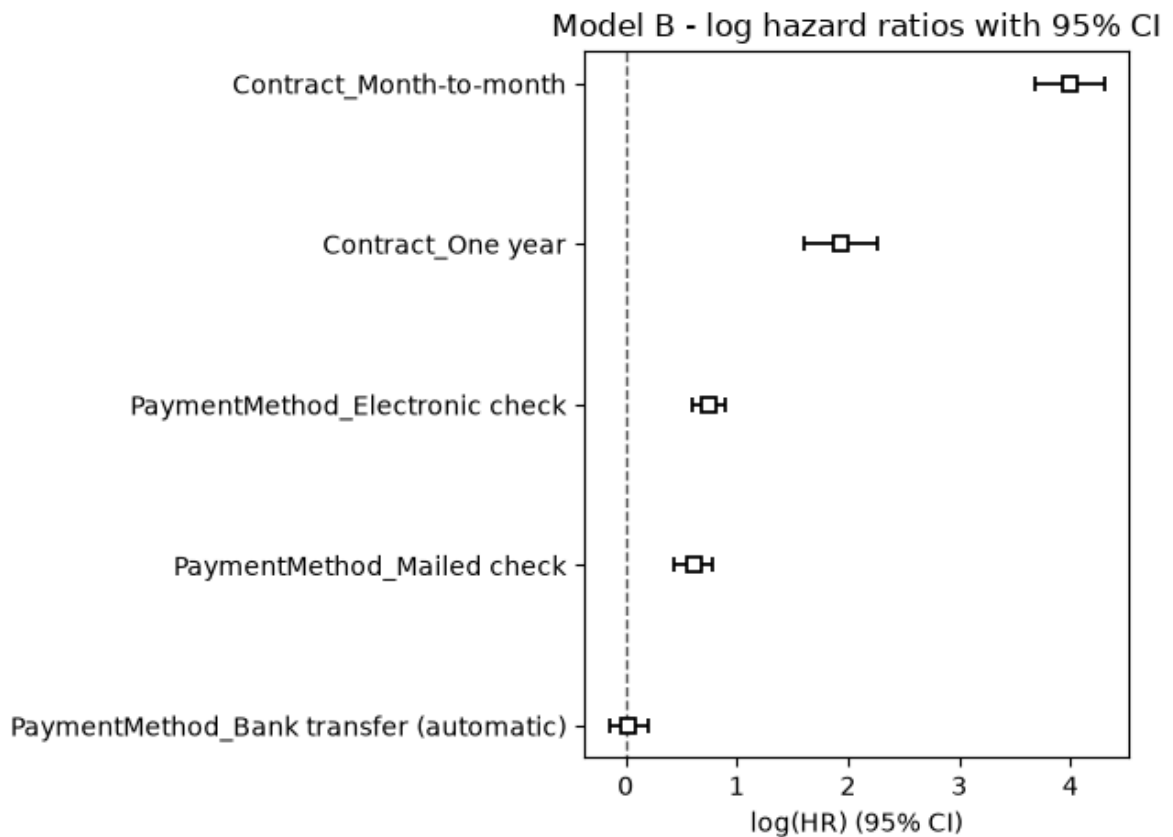
```
alone (A)"]  
comparison
```

Out[14]:

	HR alone (A)	HR adjusted (B)	shrinkage
covariate			
PaymentMethod_Bank transfer (automatic)	1.089	1.028	0.056
PaymentMethod_Electronic check	4.621	2.104	0.545
PaymentMethod_Mailed check	2.149	1.848	0.140

In [15]:

```
ax = cox_b.plot()  
ax.set_title("Model B - log hazard ratios with 95% CI")  
plt.tight_layout()  
plt.savefig("figures/cox_hazard_ratios.png", dpi=150, bbox_inches="tight")  
plt.show()
```



Does the model's own assumption hold?

Cox assumes a covariate's hazard ratio is *constant over time*. That is testable, and here it fails for some covariates - so each HR above should be read as an average effect across the 72 months, not a fixed multiplier. The mailed-check curve in section 3 shows why: it drops steeply in the first months and then flattens.

```
In [16]: proportional_hazard_test(cox_b, design,
time_transform="rank").summary[["test_statistic", "p"]]
```

```
Out[16]:
```

	test_statistic	p
Contract_Month-to-month	20.686	5.411e-06
Contract_One year	0.008	9.286e-01
PaymentMethod_Bank transfer (automatic)	0.303	5.821e-01
PaymentMethod_Electronic check	9.082	2.582e-03
PaymentMethod_Mailed check	30.595	3.180e-08

8. Minimum follow-up time

Two different questions hide under "how long do we need to watch customers".

How soon can a conclusion be drawn? Censor every customer administratively at month `t` - as if the study had stopped there - and re-run the log-rank test. The smallest `t` whose test comes back significant is the minimum follow-up that particular comparison needs.

How long do the estimates stay trustworthy? A survival curve is only as good as the number of customers still at risk behind it. Once the risk set thins out, the curve is carried by a handful of people and the confidence band widens accordingly.

```
In [17]: FOLLOWUPS = range(1, 25) # months of follow-up to try
T_all, E_all, G_all = df["T"].to_numpy(), df["E"].to_numpy(),
df["PaymentMethod"].to_numpy()

first_significant = {}
for t in FOLLOWUPS:
    censored_T = np.minimum(T_all, t)
    censored_E = ((E_all == 1) & (T_all <= t)).astype(int)
    for pair, p in pairwise_logrank_test(censored_T, G_all,
censored_E).summary["p"].items():
        if pair not in first_significant and p < 0.05:
            first_significant[pair] = (t, p)

full = pairwise_logrank_test(T_all, G_all, E_all).summary["p"]
rows = []
for pair in full.index:
    t, p = first_significant.get(pair, (np.nan, np.nan))
    rows.append({"pair": " vs ".join(pair),
                  "min follow-up (months)": t,
                  "p at that point": p,
                  "p on full 72 months": full[pair]})
```

```
pd.DataFrame(rows).sort_values("min follow-up (months)").set_index("pair")
```

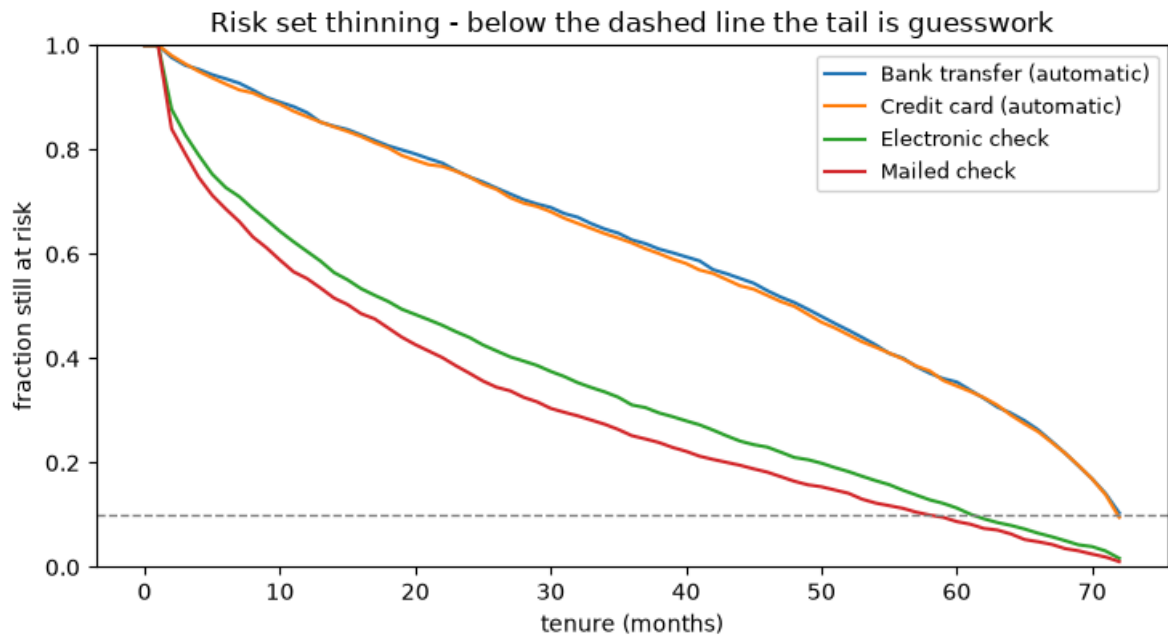
Out[17]:

	min follow-up (months)	p at that point	p on full 72 months
pair			
Bank transfer (automatic) vs Electronic check	1.0	5.673e-25	6.231e-113
Bank transfer (automatic) vs Mailed check	1.0	1.930e-19	8.905e-13
Credit card (automatic) vs Electronic check	1.0	1.765e-28	2.148e-119
Credit card (automatic) vs Mailed check	1.0	4.866e-23	8.226e-16
Electronic check vs Mailed check	2.0	5.579e-03	5.038e-35
Bank transfer (automatic) vs Credit card (automatic)	NaN	NaN	3.514e-01

```
In [18]: at_risk = pd.DataFrame({
    method: [(group["T"] >= t).mean() for t in (12, 24, 36, 48, 60, 72)]
    for method, group in df.groupby("PaymentMethod")
}, index=[f"t={t}" for t in (12, 24, 36, 48, 60, 72)]).T
at_risk.columns.name = "fraction still at risk"

fig, ax = plt.subplots(figsize=(7.5, 4.2))
for method, group in df.groupby("PaymentMethod"):
    months = np.arange(0, 73)
    ax.plot(months, [(group["T"] >= t).mean() for t in months], lw=1.5,
    label=method)
ax.axhline(0.10, color="grey", ls="--", lw=1)
ax.set(xlabel="tenure (months)", ylabel="fraction still at risk",
    title="Risk set thinning - below the dashed line the tail is guesswork",
    ylim=(0, 1))
ax.legend(fontsize=9)
plt.tight_layout()
plt.savefig("figures/at_risk_thinning.png", dpi=150, bbox_inches="tight")
plt.show()

at_risk
```



Out[18]:

	fraction still at risk	t=12	t=24	t=36	t=48	t=60	t=72
Bank transfer (automatic)		0.871	0.748	0.627	0.507	0.355	0.104
Credit card (automatic)		0.863	0.748	0.621	0.500	0.347	0.095
Electronic check		0.605	0.440	0.311	0.210	0.113	0.017
Mailed check		0.553	0.372	0.252	0.165	0.087	0.011

9. Conclusion

```
In [19]: best, worst = summary.index[-1], summary.index[0]

print(f"longest-lived group {best}")
print(f"shortest-lived group {worst}")
print()
for h in HORIZONS:
    b, w = summary.loc[best, f"RMST({h})"], summary.loc[worst, f"RMST({h})"]
    print(f"over {h:>2} months: {b:5.2f} vs {w:5.2f} -> gap {b - w:5.2f}
months ({(b - w) / h:.1%} of the window)")
print()
print(f"electronic-check hazard ratio, unadjusted
{cox_a.hazard_ratios_['PaymentMethod_Electronic check']:.2f}")
print(f"electronic-check hazard ratio, given contract
{cox_b.hazard_ratios_['PaymentMethod_Electronic check']:.2f}")
```

longest-lived group Credit card (automatic)
shortest-lived group Electronic check

over 12 months: 11.62 vs 9.86 -> gap 1.76 months (14.7% of the window)
over 24 months: 22.78 vs 17.95 -> gap 4.82 months (20.1% of the window)
over 48 months: 44.05 vs 31.32 -> gap 12.73 months (26.5% of the window)
over 72 months: 63.81 vs 41.24 -> gap 22.57 months (31.3% of the window)

electronic-check hazard ratio, unadjusted 4.62
electronic-check hazard ratio, given contract 2.10

Lifetime by payment method. Over a first year, a credit-card customer spends 11.6 of the 12 months subscribed against 9.9 for an electronic-check customer - 97% of the window versus 82%. The two automatic methods cannot be told apart at all: the log-rank test gives $p = 0.35$ on the full data, and it never falls below 0.05 at any follow-up horizon. Mailed check sits between them. Median survival time, the statistic the lecture uses, only exists for electronic check (47 months) because the other three curves never fall to 0.5, so RMST does the comparing here.

The gap grows with the window. It is 1.8 months over a year, 12.7 months over four years, 22.6 months over the full six - and as a share of the window it climbs from 15% to 31%. So a single horizon flatters or exaggerates depending on where it is cut. Twelve months is the honest one to quote because the risk set is still full there; the longer figures are real but rest on a thinning tail.

How much of that is the payment method itself. On its own, electronic check carries a hazard ratio of 4.6 against credit card. Once contract type is held constant it falls to about 2.1 - so roughly half the apparent effect was contract type wearing a payment method's clothes. What is left is still a doubled churn rate, and mailed check behaves much the same way once adjusted, which points at the manual methods themselves rather than at who chooses them.

Minimum follow-up. Every comparison the data can support is already significant after **2 months** of follow-up: four of the five pairs separate within the first month, and electronic check versus mailed check needs a second. Watching longer sharpens the estimates but changes no conclusion - except for automatic bank transfer versus credit card, which stays indistinguishable however long anyone waits. At the other end, fewer than 10% of the manual-payment customers are still at risk past 60 months, so the tail of those curves should not be quoted.